

Software engineering

Going to the web

Skill progressions by applying material:

Web development +0-2

C#+.NET +1

Singleton

Singleton
-instance: Singleton <u>+WithProperty: Singleton</u>
#Singleton() <u>+WithFunction(): Singleton</u>

```

public static void Main(string[] args)
{
    if (Singleton.WithFunction() == Singleton.WithProperty)
    {
        Console.WriteLine("Objects are the same instance");
    }
}

```

```

static Singleton instance;

```

2 references

```

protected Singleton() { } // Constructor is 'protected'

```

```

public static Singleton WithFunction()
{
    if (instance == null)
    {
        instance = new Singleton();
    }
    return instance;
}

```

1 reference

```

public static Singleton WithProperty
{
    get
    {
        if (instance == null)
        {
            instance = new Singleton();
        }
        return instance;
    }
}

```



Dependency injection

Examples

- `builder.Services.AddTransient<IExampleTransientService, ExampleTransientService>(); //Always different`
- `builder.Services.AddScoped<IExampleScopedService, ExampleScopedService>(); //Changes only with lifetime`
- `builder.Services.AddSingleton<IExampleSingletonService, ExampleSingletonService>(); //Always the same`

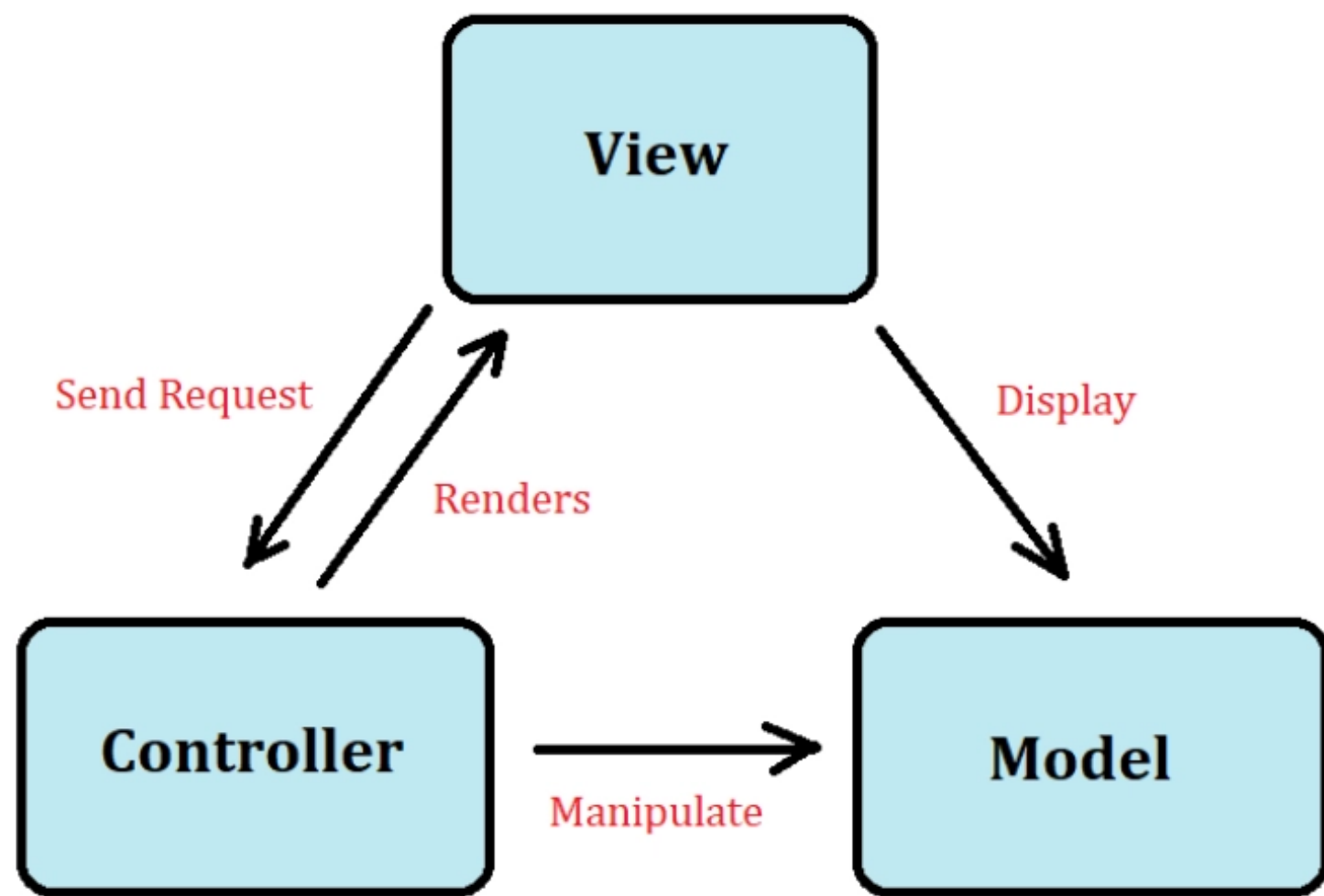
Authentication / Authorization





Web development







```
public ActionResult About()
{
    ViewBag.Message = "Yes.";
    return View();
}

public PartialViewResult Index()
{
    return PartialView("_PartialView");
}

public RedirectResult Index()
{
    return Redirect("Home/Contact");
}

public ActionResult Index()
{
    return RedirectToAction("Login", "Account");
}
```

```
public ActionResult Index()
{
    var persons = new List<Person>
    {
        new Person { Id = 1, FirstName = "Harry",
                     LastName = "Potter" },
        new Person { Id = 2, FirstName = "James",
                     LastName = "Raj" }
    };
    return Json(persons,
                 JsonRequestBehavior.AllowGet);
}

public ActionResult Index()
{
    return File("Web.Config", "text");
}
```

```

public class GuestResponse
{
    [Required(ErrorMessage = "Enter your name")]
    public string? Name { get; set; }

    [Required(ErrorMessage = "Enter your email")]
    [EmailAddress]
    public string? Email { get; set; }

    [Required]
    [Range(18, double.MaxValue, ErrorMessage = "You
must be an adult")]
    [Column(TypeName = "decimal(8, 2)")]
    public decimal Age { get; set; }
}

```

```

[HttpPost]
public ViewResult RsvpForm(GuestResponse guestResponse)
{
    if (ModelState.IsValid)
    {
        Repository.AddResponse(guestResponse);
        return View("Thanks", guestResponse);
    }
    else
    {
        return View();
    }
}

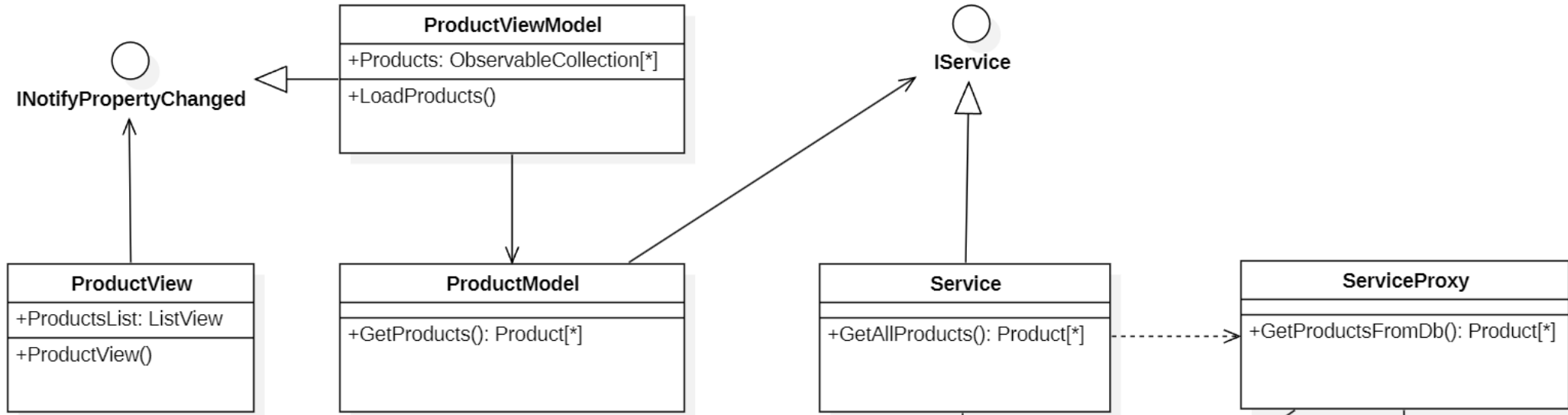
<form asp-action="RsvpForm" method="post">
    <div asp-validation-summary="All"></div>
    <label asp-for="Name">Your name:</label>
    <input asp-for="Name" />
</form>

```

```
builder.Services.AddDistributedMemoryCache();
builder.Services.AddSession(options => {
    options.IdleTimeout = TimeSpan.FromMinutes(30);
    options.Cookie.IsEssential = true;
});
app.UseSession();
```

```
@{
    Func<Product, object> row = @<tr><td>@item.Name</td><td>@item.Price</td></tr>;
}
...
<tbody>
    @foreach (Product product in Model) {
        @row(product)
    }
</tbody>
```

Application project



Server API project

